

Seolheon Kuk

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EDUCATION

KYUNG HEE UNIVERSITY

Seoul, Korea

Bachelor of Science in Information Display

Feb 2025

- Relevant Coursework: Circuit Theory, AR/VR System, TFT Reverse Engineering, Embedded Systems, Signal Processing
- Relevant Engineering Lab:
 - [Reverse Engineering of Galaxy Z Flip, Galaxy Watch, and Xiaomi Band for circuit and signal analysis.](#)
 - [Electrical Engineering: LCD panel response evaluation](#)

SKILLS

Application Software: *Advanced* - MATLAB, Unity 3D, Gateway, SmartSpice; *Intermediate* - CARLA

Programming Languages: *Advanced* - C/C++, Verilog, HDL; *Intermediate* - Python

Hardware: Oscilloscope, Probe Station, UV-Vis Spectrometer

Languages: English (Fluent), Korean (Native Speaker)

ACADEMIC RESEARCH EXPERIENCE

AI AND MOBILITY LAB

Seoul, Korea University

Full-time Researcher

Sep 2025 – Present

- Evaluated transformer-based autonomous driving workloads and benchmarked GPUs, FPGAs, and automotive SoCs to identify optimal task-accelerator mappings and synthesizing and reported prior 15% FPGA-GPU latency improvements
- Reviewed and compared key Offline RL algorithms (CQL, BCQ, MOREL, UMBRELLA) across high-, mid-, and low-level autonomous driving tasks and identified core challenges

AI SEMICONDUCTOR LAB

Suwon, Korea University

Research Assistant

Sep – Dec 2024

- Assisted in designing a 1T1R 64×64 ReRAM crossbar netlist using Gateway and the Aguirre memristor model, enabling accurate circuit-level simulation for device behavior analysis
- Developed a conductance-based weight extraction method with circuit-level readout and algorithmic validation, ensuring accurate weight identification with $\leq 5\%$ relative-error tolerance

PROFESSIONAL EXPERIENCE

ADVANCED INSTITUTE OF CONVERGENCE TECHNOLOGY

Suwon, Korea

Circuit Design Engineer Intern

July – Aug 2024

- Developed a Verilog-A-based pulsed signal generator using SmartSpice to optimize circuit performance, achieving a 96.29% reduction in simulation runtime
- Compiled weekly performance reports and tracked pending tasks for senior and PhD researchers

WORLD TRADE CENTER

Denver, CO

Industrial Engineer Intern

April – July 2023

- Maintained the backend system of Whova, an event management platform used for the World Trade Day event.
- Led agile sprints, training and peer-mentoring sessions to ensure timely delivery of startup acceleration programs

CADEMIC PROJECTS

VR GAME DESIGN AND BUILD

Fall 2024

Kyung Hee University

Seoul, Korea

- Programmed a full-stack VR game in Unity (C#/C++), implementing custom interaction logic and optimizing performance to achieve stable 90+ FPS

PUBLICATIONS

- Kuk, S., Cho, Y., & Kim, J. “Edge AI Accelerators for Transformer Workloads in Autonomous Driving: A Survey of System Constraints and Mapping Strategies” in *J-IEIE(2025)* In Press
- Kuk, S., Cho, Y., & Kim, J. “Offline Reinforcement Learning for Autonomous Driving: A Survey of Methods and Open Challenges” in *J-KICS(2025)* In Press

ACTIVITIES

- Selected Participant, SK hynix College Program (SKCE 16') Fall 2022 – Spring 2023
- Chapter president, Python Programming Club - KHU Spring 2021 – Fall 2022